

```
In [1]: # Colab + Local compatible Loader (downloads CSV if not present)
import os
import numpy as np
import pandas as pd

CANDIDATES = ['diabetes.csv', 'lab06/diabetes.csv', '/content/diabetes.csv']
data_path = next((x for x in CANDIDATES if os.path.exists(x)), None)
if data_path is None:
    url = 'https://raw.githubusercontent.com/manaal6/Introduction_To_AI_Lab_Task
    print('Downloading:', url)
    data_path = 'diabetes.csv'
    pd.read_csv(url).to_csv(data_path, index=False)
print('Using:', data_path)
df = pd.read_csv(data_path, encoding='ascii', delimiter=',')

df.head()
```

Using: diabetes.csv

```
Out[1]:
```

	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
0	148	72.0	35	169.5	33.6	0.627	50
1	85	66.0	29	102.5	26.6	0.351	31
2	183	64.0	32	169.5	23.3	0.672	32
3	89	66.0	23	94.0	28.1	0.167	21
4	137	40.0	35	168.0	43.1	2.288	33

```
In [2]: from sklearn.model_selection import train_test_split
# Prepare features and target variable
X = df[['Age', 'BMI', 'BloodPressure', 'Insulin', 'Glucose', 'DiabetesPedigreeFu
y = df['Outcome']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_

from sklearn.linear_model import LogisticRegression

# Initialize and train the Logistic regression model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
Out[2]:
```

▼ LogisticRegression ⓘ ?

► Parameters

► Fitted attributes

```
In [3]: X_test
```

```
Out[3]:
```

	Age	BMI	BloodPressure	Insulin	Glucose	DiabetesPedigreeFunction
668	43	34.0	58.0	190.0	98	0.430
324	21	35.7	75.0	102.5	112	0.148
624	21	30.8	64.0	102.5	108	0.158
690	34	24.6	80.0	102.5	107	0.856
473	50	29.9	90.0	102.5	136	0.210
...	...	...	...	...	...	...
355	49	30.4	88.0	169.5	165	0.302
534	24	33.3	56.0	56.0	77	1.251
344	57	36.8	72.0	102.5	95	0.485
296	29	28.0	70.0	360.0	146	0.337
462	39	35.3	70.0	49.0	74	0.705

154 rows × 6 columns

```
In [4]: # Weights
print(model.coef_)

# Constant / intercept
print(model.intercept_)

[[ 0.04559683  0.09394991 -0.00964773  0.00442035  0.03010364  0.48521205]]
[-9.2358725]
```

```
In [5]: from sklearn.metrics import accuracy_score, confusion_matrix, roc_curve, auc
# Predict on the test set
y_pred = model.predict(X_test)

# Calculate the accuracy score
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy Score: {accuracy:.4f}')
```

Accuracy Score: 0.7597

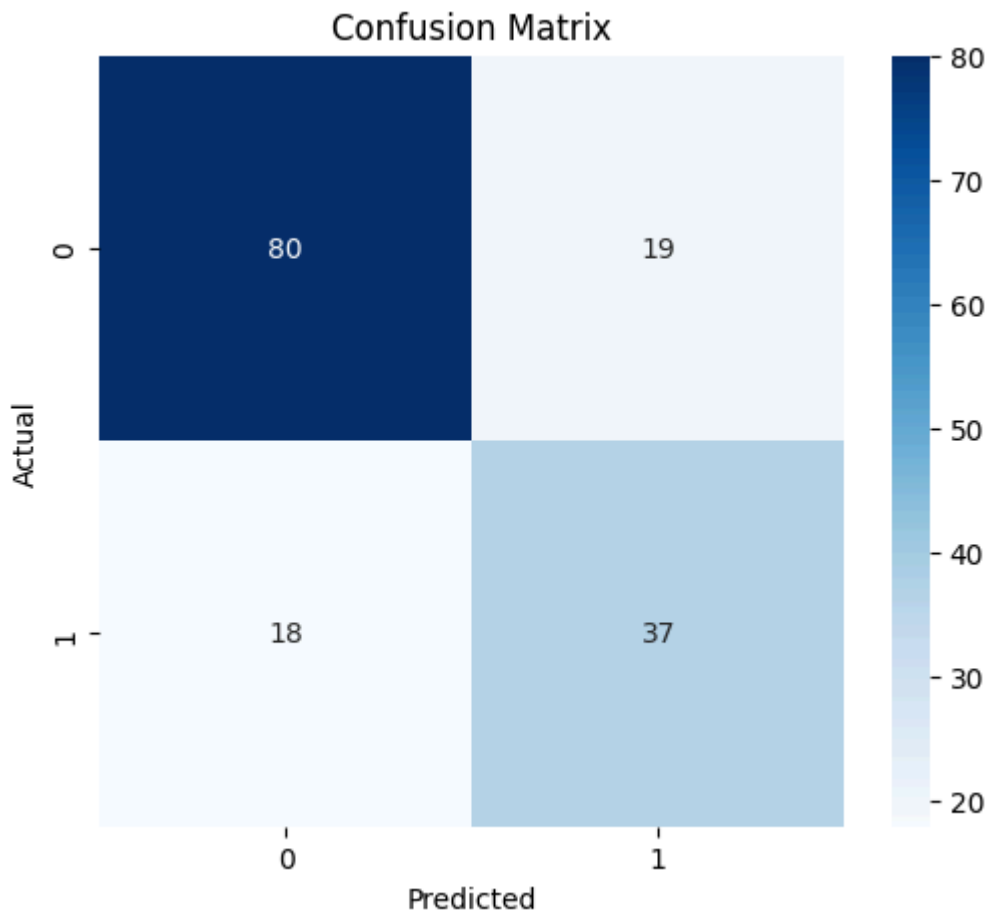
```
In [6]: y_pred
```

```
Out[6]: array([0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0,
               0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 0, 1, 0, 1, 1, 0,
               0, 1, 1, 0, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0, 1,
               0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0,
               0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 0, 0,
               0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1,
               0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0])
```

```
In [7]: %matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns

# Generate the confusion matrix and visualize using heatmap
cm = confusion_matrix(y_test, y_pred)
```

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plt.figure(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```



In [8]: 21, 35.7, 75.0, 102.5, 112, 0.148

Out[8]: (21, 35.7, 75.0, 102.5, 112, 0.148)

Part C — Prediction Dashboard (FastAPI + HTML frontend)

The trained model is served by sir's FastAPI app (`lab06/main.py`) and the user enters the six feature values in `lab06/index.html` . The page calls `POST /predict` and shows the result:

Diabetes Prediction

120
70
80
30.0
0.5
40
Predict

Result: Not Diabetic

Run it locally:

```
cd lab06
python -m uvicorn main:app --port 8000
```

then open `index.html` in the browser (it calls `http://localhost:8000/predict`).

API docs: `http://localhost:8000/docs` .

```
In [9]: # Download the trained model (.pkl) so this notebook runs anywhere (e.g. Colab),
# then Load it and verify it predicts.
import os
import pickle
import pandas as pd

PKL_URL = 'https://raw.githubusercontent.com/manaal6/Introduction_To_AI_Lab_Task
PKL_LOCAL = 'diabetes_model.pkl'

if not os.path.exists(PKL_LOCAL):
    print('Downloading:', PKL_URL)
    import urllib.request
    urllib.request.urlretrieve(PKL_URL, PKL_LOCAL)
print('Using:', PKL_LOCAL)

with open(PKL_LOCAL, 'rb') as f:
    clf = pickle.load(f)

print('Model:', type(clf).__name__)
print('Features:', list(clf.feature_names_in_))

# Sanity check: one sample prediction with feature names (no warning)
sample = pd.DataFrame([{'Age': 40, 'BMI': 30.0, 'BloodPressure': 70,
    'Insulin': 80, 'Glucose': 120, 'DiabetesPedigreeFunction': 0.5,
}])
```

```
pred = int(clf.predict(sample)[0])
prob = float(clf.predict_proba(sample)[0][1])
print(f'Predicted: {pred} ({"Diabetic" if pred else "Not Diabetic"}), P(diabetic
```

Using: diabetes_model.pkl

Model: LogisticRegression

Features: ['Age', 'BMI', 'BloodPressure', 'Insulin', 'Glucose', 'DiabetesPedigree
Function']

Predicted: 0 (Not Diabetic), P(diabetic)=0.257

C:\Program Files\Python313\Lib\site-packages\sklearn\base.py:525: InconsistentVer
sionWarning: Trying to unpickle estimator LogisticRegression from version 1.6.1 w
hen using version 1.9.0. This might lead to breaking code or invalid results. Use
at your own risk. For more info please refer to:

[https://scikit-learn.org/stable/model_persistence.html#security-maintainability-1
imitations](https://scikit-learn.org/stable/model_persistence.html#security-maintainability-1imitations)

warnings.warn(